

MIDTERM EXAM-I STUDY GUIDE

Chapter 1

- Graph Representations
- Trees

Chapter 2

- Sieve of Eratosthenes
- Euclid's Algorithm
- Asymptotic order of growth
- Analysis of Nonrecursive Algorithms
- Analysis of Loops
- Recursive Algorithms
- Backward substitutions

Chapter 3

- Selection Sort
- Bubble Sort
- Knapsack Problem
- DFS and BFS

Chapter 4

- Insertion Sort
- Topological Sorting
- Generating Permutations algorithms
 - Minimal Change
 - Johnson Trotter
- Generating subsets
 - Bit strings
- Binary Search
- Quick select-Lomuto's Partitioning

EXAM RULES

- 1- Closed notes
- 2- No cheat sheet
- 3- No calculator